



ABISHEK MANIVELAN

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@TeckTinker-Abi



@abishekmanivelan

SUMMARY

Engineering student specializing in Artificial Intelligence, Machine Learning, and IoT. Experienced in building data-driven automation systems, web applications, and mobile solutions using modern frameworks. Passionate about creating innovative, sustainable, and efficient technologies with a strong focus on solving real-world problems.

PROJECTS

1. Data-Driven Smart Agricultural Rover Prototype with IoT-Based Irrigation Control | IoT, Embedded Systems, Python Jan 2023 - Aug 2023

- Designed and developed a prototype agricultural rover for monitoring soil, environmental, and crop growth conditions.
- Integrated soil moisture and humidity sensors to collect and log real-time data into a database for future analysis and Automated irrigation control using sensor thresholds to optimize water usage.
- *Future Scope* : integration of AI/ML to analyze plant growth data for health monitoring and harvesting prediction, enabling fully automated precision agriculture.

2. Automated Aquatic Plant Collection and Biogas Production System | IoT, Embedded Systems, Python, Sustainable Energy Nov 2023 - Jan 2024

- Developed an IoT-based system for automated aquatic plant collection with integrated sensors to monitor plant quantity, water quality, and system performance, storing data for analysis.
- Automated biomass handling and biogas production processes to ensure sustainable waste management and efficient renewable energy generation.
- *Future Scope* : integration of AI/ML for aquatic plant identification, process optimization, and predictive maintenance.

3. Wireless Stethoscope for Remote Health Monitoring | IoT, AI, TensorFlow, Bluetooth, Real-Time Signal Processing Feb 2024 - Jun 2024

- Developed a wireless stethoscope system to capture and transmit heart and lung sounds for remote diagnostics.
- Applied TensorFlow-based AI models for real-time sound classification, enabling early detection of abnormalities.
- Reduced manual effort in health checks and improved diagnostic accuracy by an estimated 20–25% in testing scenarios.

4. Smart Greenhouse Automation System with Web & Mobile Integration | ESP32, IoT, Sensors, Embedded Systems, Django, Python, HTML, CSS, JavaScript, REST API, Real-Time Systems Jan 2025 - May 2025

- Developed a fully automated greenhouse using ESP32 and multi-sensor monitoring (temperature, humidity, light intensity, CO₂, soil moisture, pH) with adaptive irrigation, lighting, heating, and cooling control based on live data and crop profiles.
 - Built a Django-based real-time web dashboard for sensor visualization, actuator control, and dynamic threshold settings, with RESTful APIs enabling seamless communication between hardware and web interface.
 - *Future Scope*: Mobile integration using .NET MAUI for remote monitoring and greenhouse management (to be added soon).
 - Achieved scalable, user-friendly automation through robust hardware-software integration, demonstrated at national-level competitions.
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EDUCATION

Bachelor of Engineering in Computer Science
(Artificial Intelligence and Machine Learning)
CGPA: 7.101 (till 6th Sem)

SETHU INSTITUTE OF TECHNOLOGY
2022 - 2026

TECHNICAL SKILLS

- **Languages:** Java, C#, Python, SQL
 - **Frameworks:** .NET, HTML, CSS, Django
 - **Developer Tools:** VS Code, Visual Studio Thonny, Git Hub, Proteus Simulation, Arduino IDE
 - **Technologies :** Internet of Things (IoT), Mobile Application Development, Web Development, Machine Learning, Gen AI.
 - **Languages:** English, Tamil
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INTERNSHIPS

- Internship in robotic and IoT Trainee through E-Yantra
 - Attended a one-week workshop on Angular Framework conducted by Amizth Infotech Madurai.
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ACHIEVEMENTS

- **Awards/Activities:**
 - Internship Certificate for attending project based robotic training through E-Yantra
 - Received 1st place in state-level for project Automated Aquatic Plant Collection and Biogas Production System held at Sethu Institute of Technology. | 2023
 - Runner-up in state-level for idea presentation of project AI-Integrated Smart Agricultural Rover and Data-Driven Irrigation System held at Hindustan Institute of Technology. | 2024
 - Secured 1st place in national-level Project Expo at NEC TECH FEST 2K25, conducted by National Engineering College, for presenting Greenhouse Automation System | 2025
 - Certified with "Introduction to Industry 4.0 and Industrial IoT" (60%) from NPTEL
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AREA OF INTEREST

- Internet of Things
- LLMs
- Machine Learning
- Artificial Intelligence
- Application Developments

HOBBIES

- Exploring New Technologies
 - Solo Backpacking
 - Hiking
 - Event Planning
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PERSONAL ATTRIBUTES

- Strong problem-solving and analytical skills .
 - Quick learner and adaptable to new technologies.
 - Work effectively in a team environment.
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4. Internal Bleeding Detection and Emergency Notification System | Wearable Sensors, AI, Deep Learning, IoT, Healthcare Technology :

Jul 2024 - Sep 2024

- Used deep learning models for analyzing physiological data to detect internal bleeding with high accuracy.
- Integrated AI-powered predictive analytics and IoT for real-time emergency alerts to healthcare providers and family.
- Addressed critical health monitoring needs with a focus on rapid response and user safety.